

Vana Kirjakeele Korpus

Corpora and Language Technology Tools for Finno-Ugric Languages
(HVEE.03.021)

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Hemangi Rane

VAKK (Vana Kirjakeele Korpus)

- contains texts from the 15th to the 19th century.
- tagged and annotated.
 - 15th, 16th, and 17th centuries – manually marked by researchers
 - 18th century - auto-tagger
- statistics
 - A total of 3,016 texts
 - 3,322,112 words.
 - Out of which, 846,265 Estonian words.
 - 1,636,162 of them annotated.

Research Question

How are the long vowels represented in Old Texts, and what variations can be seen?

Long Vowels

- pronounced for an extended period
 - compared to their short vowel counterparts
- orthography
 - represented through specific orthographic conventions.
 - the use of double vowel letters: English: 'see' - long vowel /i:/
 - diacritical marks: Devnagari : इ (short), ई (long) -> हेमांगि हेमांगी
 - other linguistic symbols: macrons (ˉ), circumflex accents (^), Ancient Greek Ω

Estonian Long Vowels

- old Estonian texts used an ad hoc orthography
 - Latin
 - German
 - Finnish
- long vowels in an open syllable were written in single
 - *loomä* 'to create' was written as *loma*.
- use of grave accent (à, è).
- modern Estonian – double vowel letters.

Data Collection (1.875.119 rows)

Language =
Estonian/islands,
Estonian/Tallinn,
Estonian/Tartu

A	B	C	D	E	F
	Sõne	Märksõna	Sõnaliik	Vormiinfo	Tähe
	(Old Spelling)	(New Spelling)			
	suuta	-suuta	adv		
	–	-			
	Abelli	Aabel	s_nimi	sg.gen.	
	\Abraham/	Aabraham	s_nimi	sg.nom.	
	y[e]me	jääma	v	pers.ind.pr.pl.1.	
	[y]eemes	jääma	v	pers.ind.pr.pl.1.	
	iæb	jääma	v	pers.ind.pr.sg.3.	
	lhesu{x}sse	Jeesus	s_nimi	sg.gen.	
	kýtame	kiitma	v	pers.ind.pr.pl.1.	
	kýtada	kiitma	v	inf.	
	öseks	öö	s	sg.tr.	
	ßelle	see	pron	sg.all.	
	sehst+	see	pron	sg.el.	
	värrikeñe	üürikene	adj	sg.nom.	
	ürrikeise	üürikene	adj	sg.gen.	
	värrikeße	üürikene	adj	sg.gen.	
	wýmbse	viimne	adj	sg.gen.	
	wýmses	viimne	adj	?	

Data Cleaning : Step I (8,456 rows)

- only long vowels (*märksõna* = new spelling)
- *märksõna* = '-' and NULL ignored.
- removed special characters
- distinct rows (*märksõna* and *sõne*)
 - Examples:

	Sõne	Märksõna	Clean_Sone	Clean_Marksona	Matched_Vowel
Record #1	Kis	Kiis	Kis	Kiis	i
	kis	Kiis	kis	Kiis	i
Record #2	Kisi	Kiis	Kisi	Kiis	i
Record #3	Brudt	pruut	Brudt	pruut	u
Record #4	Bruti	pruut	Bruti	pruut	u
Record #5	Prudt	pruut	Prudt	pruut	u
	prudt	pruut	prudt	pruut	u
Record #6	Pruti	pruut	Pruti	pruut	u

Data Cleaning : Step II (459 rows)

- distinct rows based on long vowels in new and old spellings
(*märksõna_long_Vowel* and *Sone_Long_Vowel_Extract* combination)

	Sõne	Märksõna	Clean_Sone	Clean_Marksona	Matched_Vowel	Long_Vowel_Position	Context_Position	Marksona_Long_Vowel	Sone_Long_Vowel_Extract
Record #1	Elias	Eelija	Elias	Eelija	e	(0, 2)	(0, 3)	ee	el
	Eliassest	Eelija	Eliassest	Eelija	e	(0, 2)	(0, 3)	ee	el
	Eliaselle	Eelija	Eliaselle	Eelija	e	(0, 2)	(0, 3)	ee	el
	Elial	Eelija	Elial	Eelija	e	(0, 2)	(0, 3)	ee	el
	Eliase	Eelija	Eliase	Eelija	e	(0, 2)	(0, 3)	ee	el
	Elija	Eelija	Elija	Eelija	e	(0, 2)	(0, 3)	ee	el
	Eliam	Eelija	Eliam	Eelija	e	(0, 2)	(0, 3)	ee	el
	Elia	Eelija	Elia	Eelija	e	(0, 2)	(0, 3)	ee	el
	Eliast	Eelija	Eliast	Eelija	e	(0, 2)	(0, 3)	ee	el
	Eliassel	Eelija	Eliassel	Eelija	e	(0, 2)	(0, 3)	ee	el
Record #2	Helias	Eelija	Helias	Eelija	e	(0, 2)	(0, 3)	ee	he
	Helia	Eelija	Helia	Eelija	e	(0, 2)	(0, 3)	ee	he

Analytical Dataset (459 rows)

- Data analysis - manual

Clean_Sone	Clean_Mar	Matched_V	Long_Vowel_P	Context_Pc	Marksona_Long	Sone_Long_Vowel	Marksona_C	Sone_Context_Extra	results
					New Spelling	Old Spelling			
suuta	suuta	u	(1, 3)	(0, 4)	uu	uu	suut	suut	uu
Abelli	Aabel	a	(0, 2)	(0, 3)	aa	ab	aab	abe	a
Adami	Aadam	a	(0, 2)	(0, 3)	aa	ad	aad	ada	a
Ahas	Aahas	a	(0, 2)	(0, 3)	aa	ah	aah	aha	a
Achin	Aahim	a	(0, 2)	(0, 3)	aa	ac	aah	ach	a
Akisi	Aakis	a	(0, 2)	(0, 3)	aa	ak	aak	aki	a
Amosi	Aamos	a	(0, 2)	(0, 3)	aa	am	aam	amo	a
Ari	Aar	a	(0, 2)	(0, 3)	aa	ar	aar	ari	a
Aaronit	Aaron	a	(0, 2)	(0, 3)	aa	aa	aar	aar	aa
Assa	Aasa	a	(0, 2)	(0, 3)	aa	as	aas	ass	a
Abiud	Abihuud	u	(4, 6)	(3, 7)	uu	d	huud	ud	u
Ahaßuer2	Ahasveerus	e	(5, 7)	(4, 8)	ee	er	veer	uer2	e
Antiochias	Antiookia	o	(4, 6)	(3, 7)	oo	oc	iook	ioch	o
Arimathiast	Arimaatia	a	(4, 6)	(3, 7)	aa	at	maat	math	a
Pahurimist	Bahuurim	u	(3, 5)	(2, 6)	uu	ur	huur	huri	u
Bethaboras	BeetBaara	e	(1, 3)	(0, 4)	ee	et	beet	beth	e

Operational Definition

Legend	Description	Example
S1	New spelling	hooreb
S2	Old spelling	horeb
VV1	Long vowel from S1	oo
VV2	Long vowel from S2	or
V1	Vowel from V1	o
V2	Vowel V2	o

Operational Definition (contd.)

```
1
2 1. [VV1 == VV2]
3   (Here VV1 is 'marksona_long_vowel' and VV2 is 'sona_long_vowel_extract'.)
4   e.g: 'üü' == 'üü'
5
6   Action: Note VV1.
7
8 2. [VV2.contains(V1)]
9   (Here: vowel V1 is from column 'marksona_long_vowel' and VV2 is 'sona_long_vowel_extract')
10  e.g:
11   - 'üg' contains 'ü'
12   - 'gü' contains 'ü'
13   - 'üi' contains 'ü'
14
15   Action: Note V1.
16
17 3. [not VV2.contains(V1)]
18   (Here: Vowel V1 is from column 'marksona_long_vowel' and VV2 is 'sona_long_vowel_extract')
19   e.g:
20   - 'bg' does not contains 'ü'
21   - 'ag' does not contains 'ü'
22   - 'gi' does not contains 'ü'
23
24   Action:
25   - Note corresponding S1 and S2
26   - Add NV (Not a vowel) flag, for further analysis.
27   (Here 'S1' is column 'Märksöna' and S2 is column 'Söne')
28
```

Data Analysis

- apply operational definition on all records = 459 rows
- collect results = 370 rows
- record problematic cases and marked them 'NV' = 89 rows

Results (370 rows)

		Variations															
L o n g v o w e l s	aa	a	aa	e	oa	àà	åå	ho	à	a'	aâ	o	ö	ä			
	ää	ää	eh	ee	äh	eä	æ	ä	e	äe	æe	ê	a	èè	ea	æ	
	ee	ä	æ	äe	e	ê	ě	ee	eê	eê	èè	êe	ěe	êe	i	ie	ö
	ii	e	i	j	y	ÿ	yy										
	oo	e	o	ô	ö	oo	oò	oô	u								
	öö	o	ö	öö	ü												
	õõ	ee	o	ö	oö	öö											
	uu	oô	öt	u	ü	ue	uo	uu	üu	v	vw	w					
	üü	i	u	ü	üü	y	ÿ	ÿÿ									

Problematic Records (89 rows)

- Category 'NV'
 - no vowel
 - Jernsalemil = Jeruusalemm
 - kz, ks = kaas
 - does not make sense :
 - eph = jääma
 - naist =need
 - Wthendes = viies
 - wühme =viima
 - loket = jooma
 - sewde = sööma
 - Al = halleluuja - short form of halleluuja?

References

- VAKK : *vakk.ut.ee*
- Lecture slides : *'History of Estonian Language'*
- *Mati Ereht, 'Estonian Language', Linguistica Uralica Supplementary Series /Volume 1, pp 273-280, 2007.*